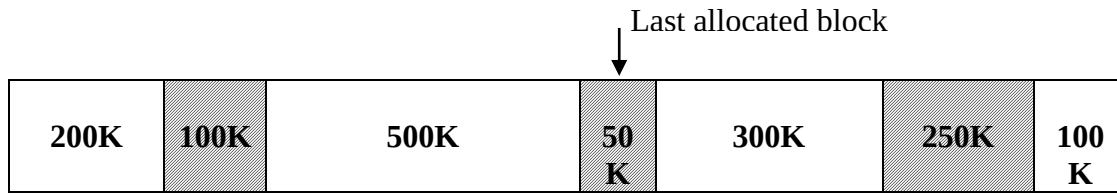


Chapter 6: Memory Management

Exercises:

1. Given memory partitions below:

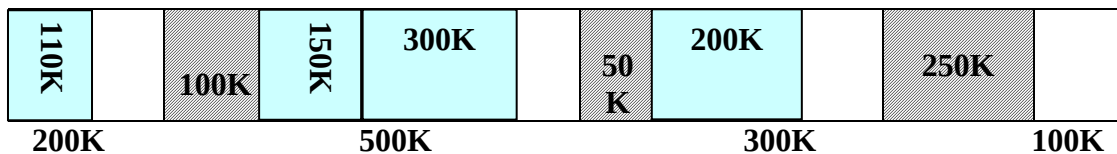


The shaded areas are allocated blocks; the white areas are free blocks.

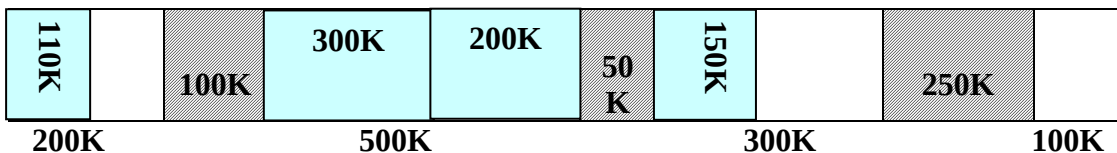
Using the following placement algorithms, show the partition allocated for the processes of size 110K, 150K, 300K and 200K (loaded in that order):

- First-fit
- Best-fit
- Next-fit

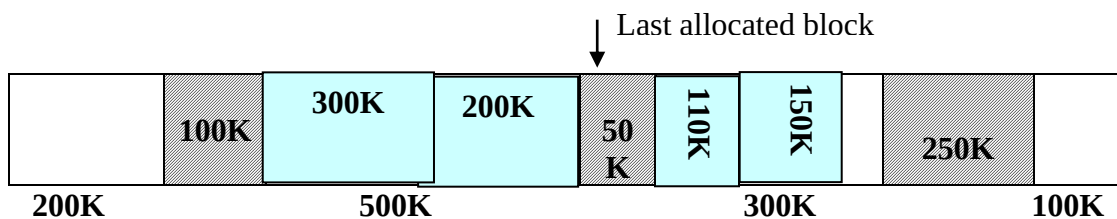
First fit:



Best fit:



Next fit:



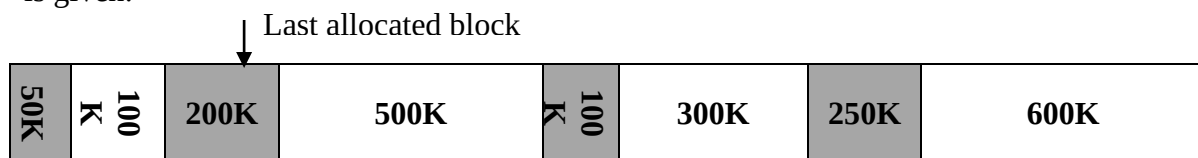
2. Consider the following segment table:

<u>Segment</u>	<u>Base</u>	<u>Length</u>
0	660	248
1	1752	422
2	222	198
3	996	604

What are the physical addresses for the following logical addresses or indicate if a segment fault occurs?

- i) (0, 198) → in Segment 0 => $660 + 198 = 858$
- ii) (1, 530) → in Segment 1 => Illegal reference
- ii) (2, 156) → in Segment 2 => $222 + 156 = 378$
- iii) (3, 444) → in Segment 3 => $996 + 444 = 1440$
- iv) (0, 222) → in Segment 0 => $660 + 222 = 882$

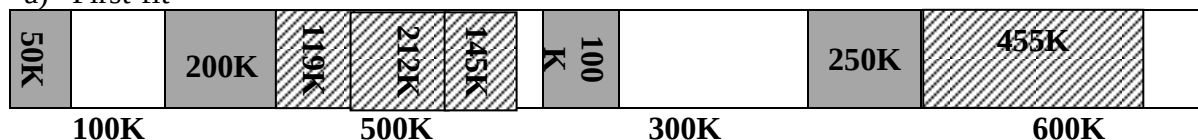
3. Dynamic partitioning scheme is being used, and the following is the main memory configuration is given:



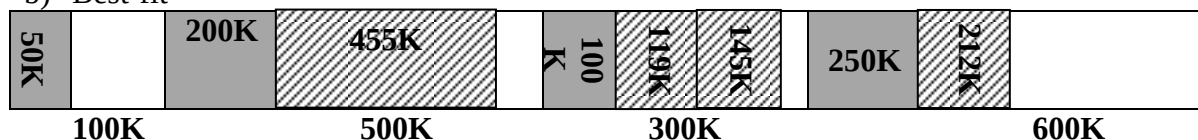
The shaded areas are allocated blocks; the white areas are free blocks.

Using the following placement algorithms, show the partition allocated for the processes of size 119K, 455K, 212K and 145K (loaded in that order):

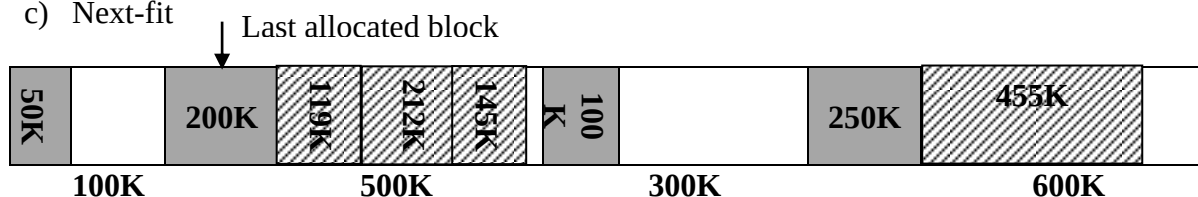
a) First-fit



b) Best-fit



c) Next-fit



4. Consider the following page table. The page size is 8 bytes each.

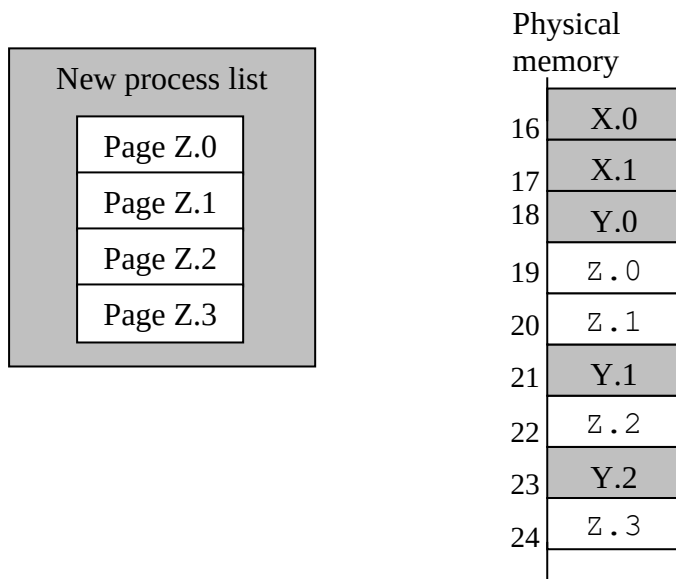
<u>Page</u>	<u>Frame</u>
0	6
1	2
2	0
3	3
4	5

What are the physical addresses for the following logical addresses?

- i) (1, 5) → in Frame 2 => $2 \times 8 + 5 = 21$
- ii) (2, 0) → in Frame 0 => $0 \times 8 + 0 = 0$
- iii) (0, 6) → in Frame 6 => $6 \times 8 + 6 = 54$
- iv) (3, 8) → in Frame 3 => Illegal reference

5. Solve the paging problem based on the information given below: pages of processes X and Y have already occupied several frames in main memory, while a new process Z is to be loaded in.

a)



b) Show the frame (in the physical memory) that is allocated to each page of the new process.

c) Draw the appropriate page-table for each process.

	X		Y		Z
0	16	0	18	0	19
1	17	1	21	1	20
		2	23	2	22
				3	24

6. Consider the following segment table:

Segment	Base	Length
0	996	248
1	2180	422
2	222	198
3	660	204
4	1752	96

a) Illustrate using diagram the places of all segments in main memory.

b) What are the physical addresses for the following logical addresses?

i) (0, 178) → in Segment 0 => $996 + 178 = 1174$

ii) (1, 256) → in Segment 1 => $2180 + 256 = 2436$

iii) (2, 130) → in Segment 2 => $222 + 130 = 352$

iv) (3, 244) → in Segment 3 => Illegal reference

v) (4, 73) → in Segment 4 => $1752 + 73 = 1825$

7. Consider a simple paging system which uses 16 bits addressing scheme. The page size is 1K each. Given a process page table as follows,

0	000101
1	000111
2	001001
3	001011
4	001100

Process Page Table

Determine the physical address of the following logical address for this scheme.

a) 000100001111110

b) 0000111111111111

Answers:

a) $\boxed{000100} \boxed{0001111110} = \boxed{001100} \boxed{0001111110}$
 Page 4 Offset Frame

b) $\boxed{000011} \boxed{1111111111} = \boxed{001011} \boxed{1111111111}$
 Page 3 Offset Frame

8. Consider a simple segment system which uses 16 bits addressing scheme. The segment size is 1K each. Given a process segment table as follows,

Segment	Length	Base
0	11110 11111	0000000010100000
1	11100 11111	0000000001111000
2	11111 00000	0000000010011010
3	10000 00000	0000000010010000
4	11111 11111	0000000010101000

Determine the physical address of the following logical address for this scheme.

a) 0000010111100000

b) 0001001110001111

Answers:

a) 0000010111100000

↑ Offset

Segment 0 → Base: 0000000010100000 +
 Offset: 010111100000

 0000011010000000

b) 0001001110001111

↑

Segment 1 → Base: 0000000001111000 +
 Offset: 001110001111

 0000010000000111

9. If a memory size is 240MB. By using the buddy partitioning scheme, shade the memory insertion for the memory space request as follows: 30MB, 50MB, 60MB, Release A, Request 40MB, Release B, Release C and Release D.

Saiz satu ingatan adalah 240MB. Dengan menggunakan skema 'buddy partitioning', tunjukkan lakaran kemasukan ruangan ingatan bagi permintaan ruangan ingatan seperti berikut: 30MB, 50MB, 60MB, pemulangan A, permintaan 40MB, pemulangan B, pemulangan C dan pemulangan D.

	240M				
Req 30M	A=30M	30M	60M	120M	
Req 50M	A=30M	30M	B=50M	120M	
Req 60M	A=30M	30M	B=50M	C=60M	60M
Release A	60M		B=50M	C=60M	60M

Req 40M	D=40M	B=50M	C=60M	60M
Release B	D=40M	60M	C=60M	60M
Release C	D=40M	60M	120M	
Release D	240M			

10. Describe a mechanism for enforcing memory protection in order to prevent a program from modifying the memory associated with other programs.

The processor could keep track of what locations are associated with each process and limit access to locations that are outside of a program's extent. Information regarding the extent of a program's memory could be maintained by using base and limit registers and by performing a check for every memory access.

11. What is the main difference between a page and a frame?

In an equal fixed-size memory partitioning, each process is also divided into small fixed-size chunks of the same size known as *pages*, could be assigned to available chunks of memory, known as *frames* or *page frames*.

12. Why does fix partitioning suffer from internal fragmentation whereas dynamic partitioning suffers from external fragmentation? When is compaction needed?

Fixed partitioning suffers from internal fragmentation because some processes may use less memory than the fixed partition size.

Dynamic partitioning suffers from external fragmentation because the memory holes left between two partitions may be too small for another process to use.

Compaction is used in a system with dynamic partitioning to place all of the partitions in a contiguous range of memory. The effect of this is to combine all the small holes into one larger hole.

13. What is the difference between a physical address and a logical address?

A physical address is an actual location in main memory whereas a logical address is a reference to a memory location independent of the current assignment of data to memory.

14. What is the difference between page, frame and segment in memory management?

Page - divide each process into the same size chunks

Frames - partition memory into small equal fixed-size chunks

Segment - divide process into segments but do not have to be of the same length

15. In memory management, blocks of data loaded into memory partitioning can cause a condition called fragmentation. Define fragmentation and explain briefly types of fragmentation that exists.

Fragmentation is a condition that occurs when processes are loaded into memory and free memory space is broken into pieces.

External fragmentation is a phenomenon associated with dynamic partitioning, and refers to the fact that a large number of small areas of main memory external to any partition accumulates.

Internal fragmentation refers to the wasted space internal to a partition due to the fact that the block of data loaded is smaller than the partition (fixed partition-page/frame).

Virtual Memory

16. Consider the following page-reference string: 2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5, 2

If the number of frames is three, fill in the frames figure below for each page replacement that would occur. Identify how many page faults that would occur for each of the following page replacement algorithms:

i. First-In-First-Out (FIFO) replacement algorithm

2	3	2	1	5	2	4	5	3	2	5	2
2	2	2	2	5	5	5	5	3	3	3	3
	3	3	3	3	2	2	2	2	2	5	5
			1	1	1	4	4	4	4	4	2
F	F		F	F	F	F		F		F	F

Page faults = 9

ii. Least-Recently-Used (LRU) replacement algorithm

2	3	2	1	5	2	4	5	3	2	5	2
2	2	2	2	2	2	2	2	3	3	3	3
	3	3	3	5	5	5	5	5	5	5	5
			1	1	1	4	4	4	2	2	2
F	F		F	F		F		F	F		

Page faults = 7

iii. Optimal replacement algorithm

2	3	2	1	5	2	4	5	3	2	5	2
2	2	2	2	2	2	4	4	4	2	2	2
	3	3	3	3	3	3	3	3	3	3	3
			1	5	5	5	5	5	5	5	5
F	F		F	F		F			F		

Page faults = 6

17. Redo the above question using:

- 5 frames
- 6 frames

18. Consider the following page-reference string: **6, 3, 2, 1, 4, 5, 6, 4, 2, 3, 2, 1, 5, 2, 1**

If the number of frames is four, fill in the frames figure below for each page replacement that would occur. Identify how many page faults that would occur for each of the following page replacement algorithms:

i. First-In-First-Out (FIFO) replacement algorithm

6	3	2	1	4	5	6	4	2	3	2	1	5	2	1
6	6	6	6	4	4	4	4	4	3	3	3	3	3	3
	3	3	3	3	5	5	5	5	5	5	1	1	1	1
		2	2	2	2	6	6	6	6	6	6	5	5	5
			1	1	1	1	1	2	2	2	2	2	2	2
x	x	x	x	x	x	x		x	x		x	x		

Page faults = 11

ii. Least-Recently-Used (LRU) replacement algorithm

6	3	2	1	4	5	6	4	2	3	2	1	5	2	1
6	6	6	6	4	4	4	4	4	4	4	4	5	5	5
	3	3	3	3	5	5	5	5	3	3	3	3	3	3
		2	2	2	2	6	6	6	6	6	1	1	1	1
			1	1	1	1	1	2	2	2	2	2	2	2
x	x	x	x	x	x	x		x	x		x	x		

Page faults = 11

iii. Optimal replacement algorithm

6	3	2	1	4	5	6	4	2	3	2	1	5	2	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

6	6	6	6	6	6	6	6	6	3	3	3	3	3	3
	3	3	3	3	5	5	5	5	5	5	5	5	5	5
		2	2	2	2	2	2	2	2	2	2	2	2	2
			1	4	4	4	4	4	4	4	1	1	1	1
x	x	x	x	x	x				x		x			

Page faults = 8

19. Redo the above question using:

- c. 5 frames
- d. 3 frames

20. Consider the following page-reference string: 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 7, 1

If the number of frames is three, fill in the frames figure below for each page replacement that would occur. Identify how many page faults that would occur for each of the following page replacement algorithms:

i. First-In-First-Out (FIFO) replacement algorithm

3	4	2	1	5	6	2	1	2	3	7	6	3	7	1
3	3	3	1	1	1	2	2	2	2	7	7	7	7	7
	4	4	4	5	5	5	1	1	1	1	6	6	6	6
		2	2	2	6	6	6	6	3	3	3	3	3	1
F	F	F	F	F	F		F		F	F	F			F

Page faults = 11

ii. Least-Recently-Used (LRU) replacement algorithm

3	4	2	1	5	6	2	1	2	3	7	6	3	7	1
3	3	3	1	1	1	2	2	2	2	2	6	6	6	1
	4	4	4	5	5	5	1	1	1	7	7	7	7	7
		2	2	2	6	6	6	6	3	3	3	3	3	3
F	F	F	F	F	F	F	F		F	F	F			F

Page faults = 12

iii. Optimal replacement algorithm

3	4	2	1	5	6	2	1	2	3	7	6	3	7	1
3	3	3	3	5	6	6	6	6	6	6	6	6	6	1
	4	4	1	1	1	1	1	1	1	7	7	7	7	7
		2	2	2	2	2	2	2	3	3	3	3	3	3
F	F	F	F	F	F				F	F				F

Page faults = 9

21. Describe the term 'replacement policy' within an operating system.

Answer:

Deals with the selection of a page in main memory to be replaced when a new page must be brought in.

22. What is the difference between simple paging and virtual memory paging?

Answer:

Simple paging: all the pages of a process must be in main memory for process to run, unless overlays are used.

Virtual memory paging: not all pages of a process need be in main memory frames for the process to run.; pages may be read in as needed.

23. When does a page fault occur? Describe what the operating system does to handle a page fault.

Answer:

A page fault occurs when a reference is mapped to a page that is not currently in main memory. When this happens, the process is suspended and the operating system takes control. If main memory is full, then a page must be replaced. Once an empty frame is found, the operating system schedules an I/O operation to bring in the page. After the page is brought in, and the page table updated, the process can be started again.

24. What is thrashing and how can an operating system take steps to avoid it?

Answer:

Thrashing occurs when a system spends more times swapping than running a process.

The operating system can avoid thrashing by reducing the number of processes that are in memory, i.e. by swapping some out.